

Total No. of Questions : 4]

SEAT No. :

PD80

[Total No. of Pages : 2

[6410]-401

T.E. (Civil) (Insem)

WASTE WATER ENGINEERING
(2019 Pattern) (Semester-II) (301012)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2 Q.3 or Q.4.
- 2) Figures to the right indicate full marks.
- 3) Use of calculator is allowed.
- 4) Assume suitable data, if necessary.

- Q1)** a) Write the necessity of pumping of the sewage. [4]
- b) Define DO deficit. Explain the Oxygen Sag curve. [5]
- c) Relate the significance of 'Self Cleansing Velocity' and 'Limiting Velocity' of sewage flow in the sewers. Calculate the minimum velocity and gradient required to transport coarse sand of 1 mm diameter through a sewer of 0.60 m diameter. Assume sewer running half full. Take specific gravity = 2.66, $\beta = 0.06$, $f = 0.02$ and $N = 0.012$. [2+4]

OR

- Q2)** a) Explain in brief self-purification of natural streams. [5]
- b) Write Streeter-Phelps equation and explain the meaning of each term. [2+3]
- c) The following data is available regarding various types of area and the corresponding impermeability factors of a town: [5]

Sr.No.	Type of catchment	% of area	Impermeability coefficient
1	Roofs	15%	0.9
2	Pavements	20%	0.8
3	Lawns & gardens	40%	0.15
4	Unpaved	15%	0.20
5	Wooded	10%	0.05

Determine the average coefficient of runoff.

If the total area of the district is 20 hectares, determine the maximum storm water flow for a rainfall intensity of 50 mm/hour having a frequency of once in five years. Use rational formula.

P.T.O.

Q3) a) Draw a flowchart for sewage treatment plant consisting primary and secondary treatment. Enlist the pollutants removed from each unit operation and process. [2+3]

b) Design a rectangular grit chamber from following data. [5]

Maximum Flow of sewage = 20 MLD

Specific gravity of grit = 2.65

Diameter of grit particle to be removed = 0.2 mm and above

Average temperature = 20°C

Flow through velocity = 0.23 m/sec

Width of grit chamber = 1.2 m

Freeboard = 0.3 m

Space of the sludge accumulation = 0.25 m

c) Design a circular settling tank unit for a primary treatment of sewage at 12 million liters per day. [5]

Assume,

i) Detention time = 2 hr

ii) Surface Loading Rate = 45000 l/m²/d

OR

Q4) a) Write the significance of velocity control devices in horizontal flow rectangular grit chambers. Explain any one Velocity control device. [2+3]

b) Design a grit chamber for the following data: [5]

i) Flow = 1650 m³/hr

ii) Settling Velocity of particle = 0.01 m/s

iii) Flow through velocity = 0.30 m/s

c) Design a primary settling tank of rectangular shape for a town having population of 150000 with a water supply of 180 liters per capita per day. Take overflow rate = 30 m³/d/m². Assume suitable data if required. [5]

